01249

Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Information Technology

Final Examination
Year 3, Semester 2 (2023)

IT3060 – Human Computer Interaction

Duration: 2 Hours

November 2023

Instructions to Candidates:

- This paper is preceded by a 10-minute reading period. The supervisor will indicate when answering may commence.
- This paper has 4 questions.
- Answer all questions in the booklet given.
- The total marks for the paper is 100.
- This paper contains 4 pages, including the cover page.
- Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.

Question 01 (20 Marks)

a) State whether you Agree or Disagree with the statements given below. Justify your answer by using the underlined phrases in user experience (UX) domain.

i. **"Sometimes when you innovate, you make mistakes. It is best to admit them quickly, and get on with improving your other innovations."- Steve Jobs**
(4 Marks)

ii. **"You've got to start with the customer experience and work backwards to the technology. You can't start with the technology and try to figure out where can I sell it." - Steve Jobs** **(4 Marks)**

b) Differentiate between UI design and UX design. **(4 Marks)**

c) Name four (04) UX design principles/factors in Human-Computer Interaction (HCI).
(2 Marks)

d) Describe what is mean by 'Context' HCI. Explain four contexts available in HCI.
(6 Marks)

Question 02 (30 Marks)

You have been assigned to design a mobile app for a new coffee shop chain called "**CofeeBrewMasters**".

a) Incorporating the design thinking model into "**CoffeeBrewMasters**" design process will be beneficial. Apply design thinking model for the above scenario and briefly describe each phase. **(10 Marks)**

b) Identify three (03) user groups for the "**CoffeeBrewMasters**" app and provide a brief description of each group. **(6 Marks)**

c) Select two (02) user groups of **"CoffeeBrewMasters"** app and create a hypothetical persona for each group. Use the below proto persona template to complete the persona.
(8 Marks)

Picture	Demographics
Behaviors	Goals & pain points

d) Ideation is the heart of the design thinking process. There are hundreds of ideation techniques. Compare and contrast "Brainstorm" and "Brainwrite" techniques.

(6 Marks)

Question 03 (35 marks)

a) **"I do not need to spend effort on prototype. I will focus on how to implement the final product"** Do you agree or disagree with this statement? Justify your reasoning.

(5 Marks)

b) Suppose you are considered as an expert in usability and user experience, and have been hired to help enhancing the user experience of a retail e-commerce platform called "**ShopVista**" that serves customers shopping for wide range of products, from electronics to clothing.

i. You are asked to gather user requirements for "**ShopVista**" application. Identify three (03) user research techniques that you would use and state the reasons for your selection. **(6 Marks)**

ii. Analyze the above scenario and address four (04) possible accessibility issues /disabilities expecting as an UX designer. Recommend design suggestions for the identified accessibility issues. **(8 Marks)**

iii. Justify the importance of applying participatory design to the above scenario.

(3 Marks)

c) You are invited for a meeting with senior management who owns the "**ShopVista**" application to demonstrate a new prototype for the e-commerce platform.

i. Briefly explain two (02) major variations of prototypes. **(5 Marks)**

ii. From the two variations in (c) (i), recommend which one allows the senior management to provide effective feedback on interactions and overall system structure. Justify your answer by providing two (02) examples for the selected choice.

(5 Marks)

d) Suggest three (03) design best practices that can be applied to the "**ShopVista**" application. **(3 Marks)**

Question 04 **(15 Marks)**

a) Mention two (02) dimensions of usability. **(2 Marks)**

b) Explain the two (02) types of usability evaluation methods and identify when they would be used in the design process. **(5 Marks)**

c) "Heuristic Evaluation is an appropriate approach for assessing the usability of an interface." Explain what is mean by Heuristic Evaluation. **(3 Marks)**

d) Explain what 'Think Aloud Evaluation Technique' is and justify why it's important as a usability engineering method. **(5 Marks)**